SECTION 25 ENVIRONMENTAL PROTECTION

NOISE CONTROL

Allowable noise limits 25.11 (1)(b)***S*** In addition to the requirement in sub-clause (1)(a), the noise level at 1m from the most affected external façade of the nearby education institutions from the construction works alone during any 30-minute period shall not exceed an equivalent sound level (Leq) of 70 db(A). The allowable equivalent sound level is reduced to 65 db(A) when the education institutions are conducting examination, including public examinations such as Diploma of Secondary School, etc. The *Contractor* shall liaise with the education institutions and/or the Examination Authority to ascertain the exact dates and times of all examination periods during the course of the Contract.

Noise mitigation measures 25.12 (6) The *Contractor* shall use movable temporary barriers whenever applicable to screen off the noisy plant from the noise sensitive receivers. The *Contractor* shall also provide purpose-built noise barrier or screens constructed of appropriate material located close to operating power mechanical equipment, such as generators and compressors, in order to reduce the noise impact to the surrounding sensitive receivers. The location and details of the barriers or screens shall be submitted to the *Supervisor* for approval before commencing works near to the educational institutes or other noise sensitive receivers.

AIR POLLUTION CONTROL

Dust suppression 25.15 (9)***S*** All dump tracks entering or leaving the Site shall be provided with mechanical covers in good service condition in accordance with the Specification given in Part B of PS Appendix L2.

(14) The *Contractor* shall ensure that dusty materials, including building debris and construction materials which are dusty by their nature, are properly covered by tarpaulin or other approved means. When dusty construction activities such as demolition work, drilling work, excavation in rock or artificial hard materials etc. are carried out in close proximity to the public, the *Contractor* shall provide dust control measures to the satisfaction of the *Supervisor*. Such measures shall include where appropriate screens or enclosures, water spraying system or the fitting of vacuum cleaning devices to pneumatic or power driven drilling, cutting and polishing machines, etc.

(15) Weigh hoppers shall be vented to suitable filters.

(16) The filter bags in the cement silo dust collector must be thoroughly shaken after cement is blown into the silo to ensure adequate dust collection for subsequent loading.

(17) For dry mix bathing, the process should be done in total by enclosure with the exhaust attached on to a fabric filter.

(18) Not used.

(19) The *Contractor* shall not install any furnace, boiler or other plant or equipment or use any fuel that might in any circumstance produce smoke or any other air pollution without the prior consent of the *Supervisor*. Unless specifically instructed by the *Supervisor*, the *Contractor* shall not light fires on site for the burning of debris or any other matter.

Smoke and exhaust control 25.16 (3) Where the public will be affected by exhaust fumes or smoke emission from any Constructional Plant or construction activities (e.g. welding) in the Site, such Constructional Plant or construction activities shall be shielded by a screen. Such screen shall be at least 1.8m in height, incombustible and shall be approved by the *Supervisor*.

(4) For Constructional Plant driven by internal combustion engines, the Contractor shall ensure that the smoke emission from the plant shall not exceed Shade 1 on the Ringelmann Chart continuously for 30 seconds at any time.

Reducing the use of volatile organic compounds 25.19A The *Contractor*'s attention is drawn to the enactment of Air Pollution Control (Volatile Organic Compounds) Regulation, which set the limits on the VOC contents of, amongst others, 51 types of paints, 7 types of printing inks and 6 major categories of consumer products. The *Contractor* shall adopt the following recommended green measures as far as practicable, or less otherwise specified in the Contract, in order to reduce the VOC emission in PS Appendix M3. The *Contractor* shall make reference to the "A Guide to Air Pollution Control (Volatile Organic Compounds) Regulation" published by Environmental Protection Department.

WASTE MANAGEMENT

Waste management – general requirements 25.20 (3) The *Contractor* shall provide EPD with one week's advance notice (with the Contractor's name and contact details) before starting to deliver the C&D waste to the landfills. The *Contractor* shall inform EPD of any subsequent changes to the disposal programme.

Reduced use of timber in temporary works 25.22 (3) In addition, the *Contractor* shall provide a summary table containing the description, justification and the estimated quantity for every work process/activity requiring the use of timber for Temporary Works construction irrespective of the quantity of timber used. A proforma of the summary table is shown in PS Appendix L5 attached.

(4) The *Contractor* shall update the summary table on the use of timber for Temporary Works construction, and submit it to the *Supervisor*'s Representative for monitoring and review by not later than the 15th day of each month, or if it is a General Holiday, the day following the General Holiday. The *Contractor* shall draw the attention of the *Supervisor*'s Representative to those work processes/activities for which the estimated quantities have been revised.

Sorting of C&D materials 25.24 (11)***S*** The *Contractor* shall establish a mechanism to record the quantities of C&D materials generated each month, using the monthly summary "Waste Flow Table" (WFT) as given at Appendix L4. The *Contractor* shall complete the monthly summary WFT, and submit it to the *Supervisor*'s Representative together with the updated sections of EMP (if any) by not later than the 15th day of each month following the month reported on, or if it is a General Holiday, the day following the General Holiday.

(12) The *Contractor* shall also submit the latest estimate of the TOTAL amount of C&D materials that are expected to be generated by the *works*. Such information should be submitted together with the monthly summary "Waste Flow Table" as required above.

(13) The *Contractor* shall record the quantities of all the recyclable materials before removal off the Site by the recycling contractors, and include the details in the WFT for submission to the *Supervisor*'s Representative pursuant to PS clause 25.24(11).

Trip ticket system 25.25**S** (1) The *Contractor* shall implement a trip ticket system (TTS) for tracking the removal of construction and demolition (C&D) materials from the Site to the disposal grounds. The non-inert portion of the C&D materials that are not recyclable shall be disposed of at West New Territories (WENT) Landfill or designated landfills as directed by the *Supervisor*.

(2) The *Contractor* shall inform the *Supervisor* of the account number of the billing account for disposal of construction waste under the Waste Disposal (Designated Waste Disposal Facility) Regulation (Cap. 354L). This is to enable the Supervisor to check the disposal records posted at the Environmental Protection Department's website. The *Supervisor* will provide the account number to the Civil Engineering and Development Department for overall monitoring of the trip ticket system, detecting and taking action to deal with malpractice such as overloading of dump trucks and improper covering of load, and compiling statistics as well as counting eligible trips for mechanical dump truck covers under the pay for safety and environment scheme pay for safety scheme.

(3) In order to make use of C&D materials generated by the Site, the *Contractor* shall use his best endeavours to identify recycling facilities or other construction sites where such materials can be used. The *Contractor* should not propose as an alternative disposal ground a private construction site which is not administered by an authorized person as defined under the Buildings Ordinance (Cap. 123) ("Authorized Person"). Where the *Contractor* has identified such a recycling facility or a construction sites which can be used as an alternative disposal ground, he shall obtain the written approval of the *Supervisor*, who will process the *Contractor*'s request expeditiously. In support of the request for such approval the *Contractor* shall provide relevant information including:

(a) a detailed description of the alternative disposal ground, including location, lot number (where appropriate), location plan and photographs of the proposed alternative disposal grounds showing the surrounding environment and land use;

(b) where the alternative disposal ground is a private construction site, a letter from each of the relevant authorities, such as Agriculture, Fisheries and Conservation Department, Lands Department and Planning Department, to comment on suitability of the site under their respective purview, and a letter from the Authorised Person of the development to confirm:

(i) the C&D materials for use in the development is acceptable;

(ii) the land/pond filling in the proposed alternative disposal ground and the use of land so formed by the C&D materials are in conformity with the statutory town plan/lease conditions; and

(iii) the *Supervisor*'s staff are allowed to enter the alternative ground to conduct inspections where necessary;

(c) where the alternative disposal is a private recycling facility, it is on the recyclers list for C&D materials recognized by EPD, as well as a letter from the operator to confirm the *Supervisor*'s staff are allowed to enter the recycling facility to conduct inspections where necessary;

(d) where the alternative disposal ground is a construction site of Government (other than a government contract quarry). Hong Kong Housing Authority or Mass Transit Railway Corporation, a written consent from the project office of the alternative disposal ground to use the C&D materials generated from the Sites;

(e) where the alternative disposal ground is a government contract quarry, a written consent from the Mines Division of Civil Engineering and Development Department to import the C&D materials generated from the Site;

(f) the estimated quantity and type of C&D materials to be used/processed in the alternative disposal ground and the approximate delivery programme, together with the name, post and specimen signature of the competent person to sign the Disposal Delivery Form. (DDF); and

(g) a system for transmitting disposal records from the alternative disposal ground to the *Supervisor*'s Representative.

(4) For illegal dumping of C&.D materials; or disposal of C&D materials at a disposal ground other than that designated in the *contract* or directed or approved by the *Supervisor* and where the

disposal ground is within any private land constitutes a major improper disposal (as defined in sub-clause 16 of this Clause) for the purposes of assessing the performance of the *Contractor*, the *Contractor* may be subject to relevant regulating and enforcement/prosecution action.

(5) The *Contractor*'s attention is drawn to *additional conditions of contract* Clause D25 on Disposal Grounds.

(6) The *Contractor* shall prepare a site management plan for implementation of the TTS for the whole *contract*. The *Contractor* shall submit within 45 days of the date of the *Client*'s letter of acceptance of the Tender the site management plan to the *Supervisor* for approval. If the *Supervisor* is of the opinion that the site management plan does not meet the requirements of this Particular Specification, he shall request the *Contractor* to revise the plan by notice in writing. The *Contractor* shall then revise the plan and resubmit it within 7 days of the date of the notice. The *Contractor* shall review the site management plan on a monthly basis. The updated section of the plan (if any) shall be submitted to the *Supervisor* for approval. The plan shall include the following details:

(a) Site organization and staff duties

A site organisational chart showing the manpower resources and duties of each staff for implementation of the TTS. The *Contractor*:

(i) shall appoint a senior staff member (with at least two years experience in site management) fully responsible for implementing and overseeing the operation of the TTS;

(ii) shall appoint designated person(s) to fill in and sign Part 1 of the Daily Record Summary (DRS) properly before departure of the truck. A sample of the DRS is given at Appendix M7; and

(iii) shall appoint experienced person(s) to man each exit from the Site for the purpose of ensuring that every truck carrying C&D materials leaving the Site A bears a duly completed CHIT/DDF. The CHIT shall be used for disposal of C&D materials at a prescribed facility as defined under the Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N) (hereinafter referred to as "prescribed facility"). A sample of the CHIT is given in Appendix M5. The DDF shall be used for disposal of C&D materials at other disposal grounds as designated in the *contract* or as directed by the *Supervisor*, or recycling facilities/construction sites proposed by the *Contractor* and approved by the *Supervisor*. A sample of the DDF is given in Appendix M6. The person(s) who man the exit(s) shall record the CHIT/DDF no., the vehicle registration mark and the departure time of every truck

carrying C&D materials leaving the Site.

(b) Disposal Programme

The *Contractor* shall prepare a monthly programme [or bi-weekly programme as determined by the *Supervisor* for disposal of C&D materials off the Site, and indicate the estimated quantities, types of the C&D materials and corresponding disposal grounds. The *Contractor* shall update the programme on a monthly basis and submit it to the *Supervisor* for information by 15th day of each month or the next working day if it is a public holiday, Sunday or Saturday, or a later date as agreed by the *Supervisor*.

(c) Site Procedures

The *Contractor* shall establish site procedures to ensure that each truckload of C&D materials leaving the Sites will bear a duly completed CHIT/DDF and that Part 1 of the DRS has been filled in and signed properly before departure of the truck. The *Contractor* shall also establish a mechanism to ensure timely retrieval of the CHIT/DDF and/or receipt from the disposal grounds where irregularities are observed.

(d) Surveillance

The *Contractor* shall establish a surveillance system within the Sited and at any alternative disposal grounds to check that the disposal activities comply with the requirements as set out in the Particular Specification.

(e) Recording System

The *Contractor* shall maintain a comprehensive register of the CHIT/DDF issued, and make it available for inspection by *Supervisor*'s Representative upon request.

(f) Control Measures to Track Internal Movement of Materials

Where trucks need to exit and re-enter the Site for delivery of C&D materials generated by the Site, the *Contractor* shall devise control measures to ensure that the C&D materials are not disposed of outside the Site in breach of the *contract*.

(g) Not used.

(7) Recyclable materials such as metal, paper, plastics and milled bituminous materials etc., which have been sorted on the Site for the purposes of recycling, shall not be considered as C&D materials for the purposes of the *contract*. For such sorted recyclable materials, the *Contractor* shall devise appropriate control measures to ensure (that the materials are delivered to a proper recycling outlet for processing. The *Contractor* shall highlight such control measures in the environmental management plan pursuant to ETWB TCW No. 19/2005 if applicable.

(8) The *Contractor* shall obtain the approval of the *Supervisor* for the site management plan before disposing any C&D materials from the Site.

(9) The *Contractor* shall write to all truck drivers engaged for removal of C&D materials from the Site and draw their attention to the following particular points:

(a) Each truck carrying C&D materials leaving the Site for a disposal ground must bear a duly completed CHIT/ DDF, irrespective of the location and nature of the disposal ground; and

(b) The C&D materials must be disposed of at the disposal grounds stipulated in the *contract* or directed by the *Supervisor* or alternative disposal grounds approved by the *Supervisor*.

(10) The procedures for implementation of the TTS are as follows:

(a) For each truckload of C&D materials leaving the Site, the *Contractor*'s truck driver must bear a duly completed CHIT/ DDF.

(b) The *Contractor* shall maintain a daily record of disposal of C&D materials from the Site including CHIT/ DDF numbers, vehicle registration marks, driver's particulars, approximate volume, C&D materials type, designated disposal ground, departure time from the Site, actual disposal ground and arrival time at disposal ground. The *Contractor* shall complete Part 1 of the JJRS in duplicate and inform the *Supervisor*'s staff before departure of the vehicle. The *Supervisor*'s staff shall sign Part I of the DRS before departure of the trucks, or to suit site operations at other time to be agreed between the *Supervisor*'s Representative and the *Contractor*.

(c) The truck shall proceed to the disposal ground as stipulated in the *contract* or directed by the *Supervisor* or alternative disposal ground approved by the *Supervisor*. The *Contractor*'s truck driver shall present the CHIT/DDF to the operator of the disposal ground. For a prescribed facility, if the C&D materials accord with the acceptance criteria, disposal of the materials will be permitted and the facility operator will give the *Contractor*'s truck driver a Transaction Record Slip and stamp the CHIT.

(d) For disposal ground other than prescribed facilities, the *Contractor* shall ensure that the DDF is signed off by a competent person as agreed by the *Supervisor* at the disposal ground to confirm completion of each trip. The *Contractor* shall also maintain a daily record with details of each disposal trip from the Site to the disposal ground. The *Contractor* shall complete Part 2 of the DRS form for submission to the *Supervisor*'s Representative within 3

working days after the date of disposal.

(e) Where an irregularity is observed or where requested by the *Supervisor*'s Representative under special circumstances (e.g. a CHIT/ DDF has been issued but there is no disposal record at the disposal ground), the *Contractor* shall submit to the *Supervisor*'s Representative within 5 working days after the recorded date of disposal the supporting evidence such as duly stamped CHIT/DDF and/or the Transaction Record Slip (where relevant) to confirm proper completion of the delivery trips in question, or within 2 working days after the *Supervisor*'s Representative has requested for such evidence, whichever is later A fax copy of the CHIT/DDF or Transaction Record Slip is acceptable, unless otherwise directed by the *Supervisor*.

(11) The following items shall be included in the agenda for discussion at every Site Safety and Environmental Management Committee meeting, and Site Safety and Environmental Committee meeting, or other established channels for performance monitoring as agreed by the *Supervisor*'s Representative:

(a) review the site management plan and implementation of the TTS, and identify areas for improvement;

(b) audit the quantity of C&D materials removed from the Site (based on the DRS and survey records) against the quantities of C&D materials delivered to the disposal ground designated in the *contract* (e.g. based on EPD website and directed or approved by the *Supervisor*;

(c) review incidents of non-compliance and discuss the necessary follow-up actions; and

(d) monitor the follow-up action on defects and deficiencies identified.

(12) Where C&D materials from the Site have been dumped at a place other than that designated in the *contract* or directed or approved by the *Supervisor*, the *Contractor* shall at his own cost undertake the following remedial action:

(a) remove the dumped C&D materials from the unauthorised disposal ground to a disposal ground designated in the *contract* or directed or approved by the *Supervisor* to his satisfaction; and

(b) reinstate the unauthorised disposal ground to the condition before dumping of the C&D materials, or a condition considered satisfactory by the Authority as required under the relevant legislation where appropriate.

(13) Where the unauthorized disposal ground is a private property, the *Contractor* shall be responsible for obtaining the landowner's consent before removal of the dumped C&D materials.

(14) Should the *Contractor* fail to remove the C&D materials from the unauthorised disposal ground or fail to reinstate the unauthorised disposal ground as set out in sub-Clause 12 of this Clause, the *Client* may in accordance with *additional conditions of contract* Clause B4 instruct another contractor to perform the work and the *Client* shall be entitled to recover such costs from the *Contractor*, and may but shall not be bound to deduct such costs either in whole or in part, in accordance with the provisions of *additional conditions of contract* Clause B3.

(15) The following shall constitute an "improper disposal":

(a) *Contractor*'s truck loaded with C&D materials having left the Site without a CHIT/DDF or the daily record summary not completed in accordance with this Particular Specification;

(b) Disposal of C&D materials at a disposal ground other than that designated in the *contract* or directed or approved by the *Supervisor*; or

(c) The *Contractor* fails or is unable to produce the stamped CHIT/DDF or the Transaction Record Slip (where relevant) in accordance with Clause 10(1) of this Particular Specification.

(16) The following shall constitute a "major improper disposal".

(a) Illegal dumping of C&D materials; or

(b) Disposal of C&D materials at a disposal ground other than that designated in the *contract* or directed or approved by the *Supervisor*, and where the disposal ground is within any private land. For the purpose of this Clause, "private land" has the same meaning as it has in the Mining Ordinance (Cap. 285).

(17) The *Contractor* shall also note that the *Client* takes a very serious view of any non-compliance with the TTS requirements. Without prejudice to other regulating action which may be taken against the *Contractor*, the *Contractor*¡¦ performance in implementing the TTS will be fully reflected in the Report on the *Contractor*'s Performance.

(18) The *Contractor* shall comply with the provisions of the Personal Data (Privacy) Ordinance (Cap, 486) in relation to personal data collected under the TTS.

ENERGY EFFICIENCY LABELLING SCHEME

Energy Efficiency Labelling Scheme

25.35 To make it easier for the public to choose energy efficient products, EMSD operates an Energy Efficiency Labelling Scheme for appliances and equipment used both at home and office as well as for vehicles. The scheme aims to save energy by informing potential customers of the product's level of energy consumption and efficiency rating, so that buyers can take these factors into consideration when making their

purchasing decision. As a part of the environmental protection requirement, the *Contractor* shall comply with the highest energy efficiency standard or implement the suggested measures as far as it is practicable, or unless it is otherwise specified in the Contract, during the course of carrying out the *works* or comply with the *Contractor*'s obligation under the Contract. The *Contractor* will be monitored based on the following publications published by EMSD.

(a) Guide for Energy Efficient Green Office Equipment;

(b) Hong Kong Energy Efficiency Labelling Schemes for Household Appliances - Room Coolers, Refrigerators, Washing Machines, Compact Fluorescent Lamps, Electric Clothes Dryers, Household Electric Storage Water heaters, Rice Cookers, Dehumidifiers and Televisions.

(c) Hong Kong Energy Efficiency Labelling Scheme for Office Equipment;

(d) Hong Kong Energy Efficiency Labelling Scheme for Petrol Passenger Cars;

(e) Save Fuel, Save the Environment;

(f) Fuel Economy Calculator;

(g) Scheme Documents;

(h) Ideal Electrical Appliances with Energy Label;

(i) Guidelines on Application for Voluntary Energy Efficiency Labelling Scheme;

(j) Choose Dehumidifier with Energy Label can save;

(k) How to Choose Household Refrigerators and Room Coolers.